

Temperature Conversion

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

The PPU Weather Station receives temperature readings from different countries. Some countries use Celsius (C), while others use Fahrenheit (F). To help visitors understand the measurements, the station automatically converts temperatures between the two scales.

If the input **type is C**, convert Celsius to Fahrenheit using:

$$F = (9 * C) / 5 + 32$$

If the input **type is F**, convert Fahrenheit to Celsius using:

$$C = (5 * (F - 32)) / 9$$

Given the temperature type and its value, print the converted temperature.

Print the converted temperature with exactly 6 digits after the decimal point.

Input

The first line contains a character t (**C or F**).

The second line contains a real number x ($-1000.0 \leq x \leq 1000.0$).

If $t = \text{C}$, then x represents a temperature in Celsius.

If $t = \text{F}$, then x represents a temperature in Fahrenheit.

Output

Print the converted temperature.

Examples

standard input	standard output
F 10	-12.222222
C 10	50.000000